

SOCAMES 6 Design Alert 1

Problem

During a recent construction activity of cable upgrades to an existing telecommunication room (TR) at an Initial Operating Capability (IOC) site, it was discovered during cable testing procedures that some of the cables being upgraded were greater in length to the end user device than standards allows.

Background

ANSI/TIA-568.D, "Balanced Twisted-Pair Telecommunications Cabling and Components" (Sep 2018) version of industry standards for telecommunication room fixed cabling installation, limits the total length of a network Unshielded Twisted Pair (UTP) cable segment to 100 meters. The 100 meters is broken down as 10 meters for patch cables and 90 meters for total fixed cabling. Cable length requires a three-dimensional assessment to ensure that the fixed cables being installed do not exceed the specified UTP cable segment of 90 meters of fixed cable length.

There are several scenarios that require an assessment to occur to ensure that a telecommunication rooms UTP cable segments do not exceed 90 meters.

Scenario 1: The installation of a new telecommunication room to serve an area.

Scenario 2: The relocation of an existing telecommunication room to serve an area.

Scenario 3: The installation of new cabling upgrades in a telecommunication room that will remain in place to serve an area.

Scenario 4: An existing telecommunication room that will not receive any cabling upgrades.

Solution

To ensure that an assessment is performed for each of the four telecommunication room scenarios listed above, our office in conjunction with the SD-DCIE team is providing a recommended three-dimensional estimation approach that can be shared with your respective facility leads and design team to verify all telecommunication rooms do not exceed the specified UTP cable segment of 90 meters.

Step 1: Ensure you have an accurate floor plan that depicts the locations of the telecommunication rooms serving the facility.

Step 2: Utilizing AutoCAD or rule, center on the location of the telecommunication room being assessed. From the center of the room project out lines, 80 meters in length from that point, in the plan north, south, east and west respectively.

Step 3: At the end each of the 80 meters lines created, connect a line 45 degrees from that point out to the next end point creating a diamond.

Step 4: Validate that all areas served by that telecommunication room fall within the parameters of the diamond overlay.

Step 5: If the area served by that telecommunication room is within the diamond overlay then the distance, as provided by the estimation tool, is adequate for that telecommunication room.

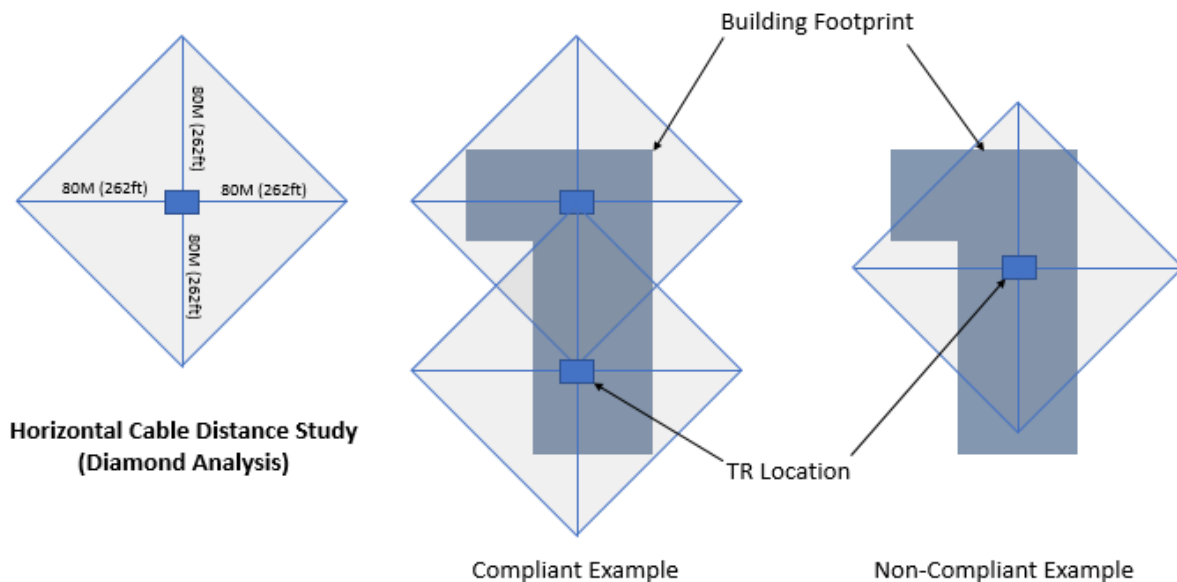
Step 6: If the area served by that telecommunication room is outside the diamond overlay then the distance, as provided by the estimation tool, is inadequate for that telecommunication room and further investigation is required by electronically testing the cable lengths to see if they pass or physical measurement. Any UTP cabling that does not pass testing protocols and is too long will need to have further design considerations to remedy the distance issue.

Notes

Since it is necessary to account for vertical cable distance, installation of slack for bypassing obstructions, cable management, bend radii and other in-situ obstacles 80 meters is utilized instead of the 90 meters acceptable length. This is to provide a realistic estimation of coverage of the representative telecommunication room.

This estimation tool is a recommended approach to validate fixed cable lengths meet the standards. Designers and installers remain responsible for ensuring actual cable distances do not exceed the limits in the standards.

Below is a diagram of an example demonstrating the use of the estimation tool.



HEFP SEP (SOCAMES 6) Design Alert 2

Problem

VA OEHRM previously required an on-site or near campus Help Desk during Go-Live. To better serve facilities during Go-Live, the Help Desk will now be remote and supported out of Kansas City.

Background

VHA HEFP SEP provided guidance in the *Electronic Health Record Modernization Training and Support Space, HEFP SEP Space Planning Job Aid*, 11/10/2020 revised 1/27/2021, that required 1,800 square feet of space for the Help Desk as part of the Command Center.

Solution

The *Electronic Health Record Modernization Training and Support Space, HEFP SEP Space Planning Job Aid*, has been updated to reflect the new requirements. The requirement for Help Desk space within the Go-Live Command Center was deleted, therefore, space is not needed for this activity.

Notes

Please use this link, [Electronic Health Record Modernization Training and Support Space, HEFP SEP Space Planning Job Aid](#), 11/10/2020 revised 3/16/2021 to find the most up-to-date information.

HEFP SEP (SOCAMES 6) Design Alert 3

Problem

VA OEHRM previously planned for 3.5 classes per end user and either 8 seats per class with social distancing or 15 seats per class without social distancing. However, during the recent VHA EHRM Strategic Review, it was determined that each end user shall have 5 classes and that there will be either 8 seats per class with social distancing or 12 seats per class without social distancing. The changes described result in a larger space need.

Background

VHA HEFP SEP provided guidance in the *Electronic Health Record Modernization Training and Support Space, HEFP SEP Space Planning Job Aid*, 11/10/2020 revised 1/27/2021, that required 3.5 classes per end user and either 8 seats per class with social distancing or 15 seats per class without social distancing. VA OEHRM and VHA determined that more class time per end user and a decrease in the overall class size are needed.

Solution

The *Electronic Health Record Modernization Training and Support Space, HEFP SEP Space Planning Job Aid*, has been updated to reflect the new requirements. The following classroom calculations were revised.

Previous classroom calculations:

Standard Room Calculations

VAMC Total rooms = Number of end users x **3.5 classes per end user / 15 seats per class** / 8 weeks of EUT / 5 days of training per week

CBOC Total rooms = Number of end users x **3.5 classes per end user / 15 seats per class** / 8 weeks of EUT / 5 days of training per week.

Social Distanced Room Calculations

VAMC Total rooms = Number of end users x 3.5 classes per end user / 8 seats per class / 8 weeks of EUT / 5 days of training per week

CBOC Total rooms = Number of end users x 3.5 classes per end user / 8 seats per class / 8 weeks of EUT / 5 days of training per week.

New classroom calculations:

Standard Room Calculations

VAMC Total rooms = Number of end users x **5 classes per end user / 12 seats per class** / 8 weeks of EUT / 5 days of training per week

CBOC Total rooms = Number of end users x **5 classes per end user / 12 seats per class** / 5 weeks of EUT / 5 days of training per week.

Social Distanced Room Calculations

VAMC Total rooms = Number of end users x 5 classes per end user / 8 seats per class / 8 weeks of EUT / 5 days of training per week

CBOC Total rooms = Number of end users x 5 classes per end user / 8 seats per class / 5 weeks of EUT / 5 days of training per week.

Notes

Please use this link, [Electronic Health Record Modernization Training and Support Space, HEFP SEP Space Planning Job Aid](#), 11/10/2020 revised 4/20/2021 to find the most up-to-date information.

HEFP SEP (SOCAMES 6) Design Alert 4

Problem

Clarification to the field is needed regarding the design of space for Telecommunications Rooms (TRs) in existing facilities for the Electronic Health Record Modernization (EHRM) initiative. The design reference, *Infrastructure Standards for Telecommunications Spaces – Version 3.0* requires 170 sq ft TRs for clinical spaces. However, due to existing space and structural constraints, HEFP SEP requires, at a minimum, only 100 sq ft TRs for clinical spaces in **existing** buildings.

Background

The *Infrastructure Standards for Telecommunications Spaces – Version 3.0* was developed by the Solution Delivery-Data Center and Infrastructure Engineering (OIT SD-DCIE) team and published to the TIL August 21, 2020. HEFP SEP identifies the following references as standards for all Information Technology (IT) physical infrastructure design and construction projects: *Infrastructure Standards for Telecommunications Spaces – Version 3.0* and *VA OEHRM Site Infrastructure and End User Device Requirements*. These references are discussed and provided to each station during and after their scheduled scoping session to identify the work needed to modernize IT infrastructure across the enterprise.

During the site scoping sessions, HEFP SEP uses the Generic Floor Plan for TRs (Two Rack Requirement), detail 3, page 155 in Appendix B-7 of the *Infrastructure Standards for Telecommunications Spaces – Version 3.0* to communicate and coordinate space requirements for **existing** TRs to support EHRM. The baseline TR template is a 10 ft x 10 ft (100 sq ft) generic floor plan with two 2 ft x 4 ft racks to house the IT infrastructure.

Infrastructure Standards for Telecommunications Spaces – Version 3.0 specifies that a clinical TR should have a generic floor plan layout of 170 sq ft to support four 2 ft x 4 ft racks. This specification for clinical TRs has created a conflict with the standardized baseline TR of 100 sq ft that HEFP SEP works with the sites to achieve.

HEFP SEP discussed the conflict in TR size requirements with the OIT SD-DCIE team. HEFP SEP identified that due to VAMC/CBOC space deficits, **existing** building footprints, and the number of TRs, it is difficult to meet the 170 sq ft space requirement without a detrimental impact to healthcare operations. Furthermore, based on the average number of end users associated with each TR, most TRs only house 2 racks and require 100 sq ft of space. HEFP SEP and OIT SD-DCIE agreed that the 100 sq ft baseline standard was the practical approach in **existing** buildings.

Solution

All EHRM projects that require TR work in **existing** spaces shall minimally use the Generic Floor Plan for TRs (Two Rack Requirement), detail 3, page 155 in Appendix B-7 of the *Infrastructure Standards for Telecommunications Spaces – Version 3.0* which requires a 10 ft x 10 ft space to support two 2 ft x 4 ft racks. Room dimensions shall be increased by 20 sq ft for any additional 2 ft x 4 ft racks.

All new construction of clinical spaces should adhere to the standards as written.

Notes

Please use this link, <https://www.cfm.va.gov/til/dguide/OIT-InfrastrucStdTelecomSpaces.pdf> to locate the current version of Infrastructure Standard for Telecommunication Spaces – Version 3.0 published August 21, 2020.

HEFP SEP Design Alert 5

Problem

Clarification is required regarding the need to replace or install new fire suppression systems in the data center during design and construction of Electronic Health Record Modernization (EHRM) facility infrastructure projects and the need for a third-party fire engineer review of HEFP SEP EHRM Infrastructure Upgrades Projects.

Background

On June 1, 2021 the new [Fire Protection Design Manual](#) was published and posted to the Technical Information Library (TIL). The manual contains fire protection engineering design for Department of Veterans Affairs facilities. EHRM facility infrastructure projects, and specifically work conducted in the Data Centers, may require an upgrade or replacement of the existing fire suppression system in accordance with NFPA 75 as stated in the [Fire Protection Design Manual](#) Paragraph 3.7.

Solution

VHA field engineers are required to ensure that the Architect-Engineer designer of record applies all applicable NFPA 75 requirements for work needed in the Data Center. Field engineers should use the Healthcare Environment and Facilities Programs (HEFP) Fire Code Plan Review Contract to review all EHRM Infrastructure Upgrades and Data Center projects. The fire code plan review should focus on compliance with the newly published Fire Design Manual dated June 1, 2021 and ensure that all associated NFPA fire and life safety codes are indicated in the designs.

Guidance to the Architect-Engineer is as follows:

For EHRM Infrastructure Upgrade and Data Center projects, the work conducted in the Data Centers (which will upgrade spaces to an ANSI/TIA-942-B, Rating 3) generally requires compliance with all requirements of NFPA 75. The most design intensive of these requirements is the addition of a gaseous fire suppression system as described in the [Fire Protection Design Manual](#), Paragraph 3.7. Compliance with NFPA 75 is also likely to involve, but not necessarily be limited to, these additional retrofit requirements:

- Revise the sprinkler protection in the computer room to a dedicated sprinkler zone.
- Ensure a minimum 1-hour boundary around the computer room.

- Combination smoke/fire dampers at the duct penetrations through the 1-hour boundary of the computer room.
- Electrical power off button(s) at computer room exit(s).
- HVAC power off button(s) at computer room(s).

Notes

Please use this link, [Fire Code Plan Reviews | Healthcare Environment and Facilities Programs \(va.gov\)](#) to request third-party fire engineer review for your EHRM upgrades projects.

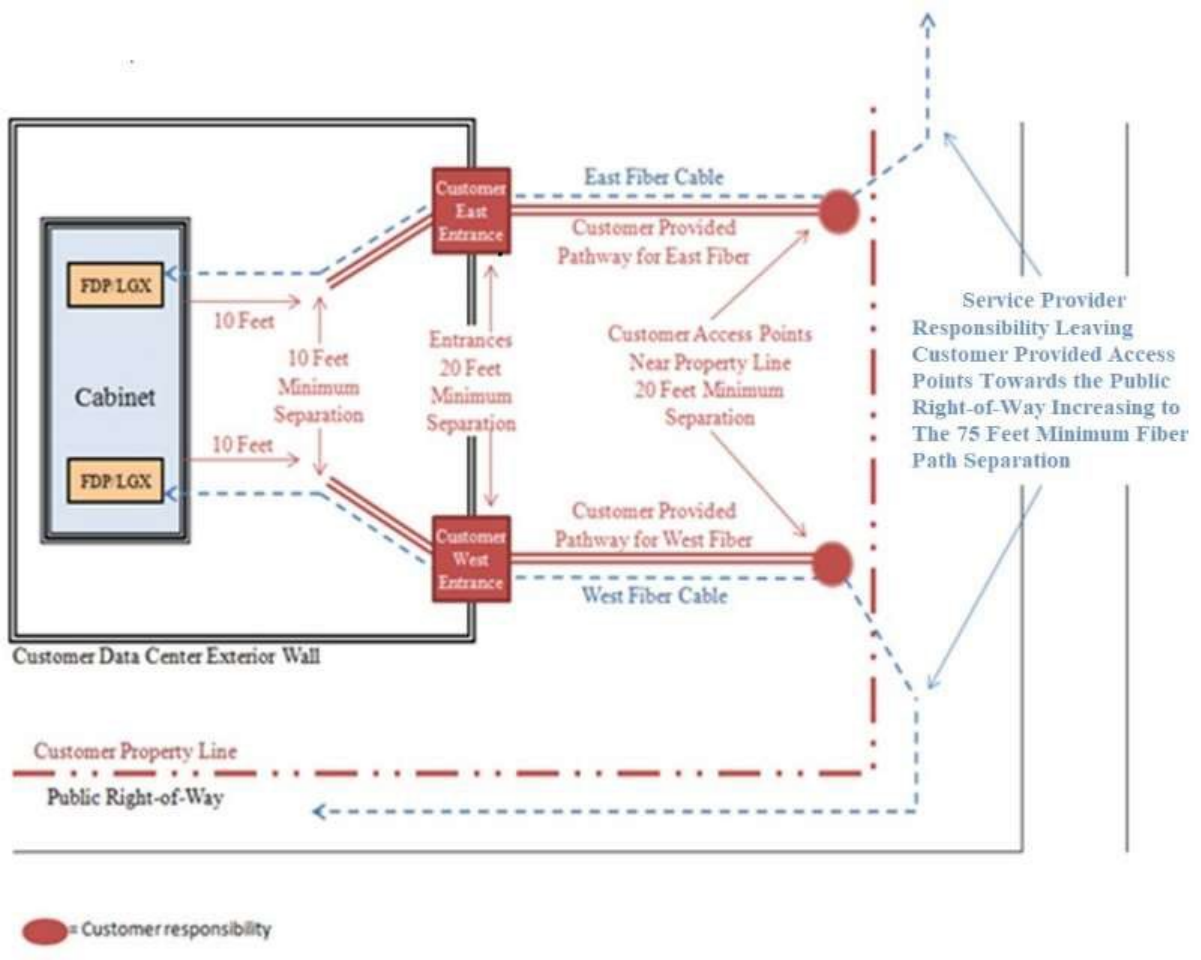
HEFP SEP (SOCAMES 6) Design Alert 6

Problem

HEFP SEP is working with OIT to create a process for installation of the additional geographically diverse path for the Wide Area Network (WAN) to ensure compliance with the VA OERHM Site Infrastructure Requirements (SIEUDR), version 2.0, paragraph 2.1.1.2.B (see below). However, HEFP SEP determined that there is currently no national OIT or OEHRM process to coordinate this work with the OIT contracted providers.

2.1.1.2 VAMC Sites

- B. Facility shall have end-to-end, geographically diverse and carrier-diverse circuits with minimum fiber pathways separation: service provider public rights-of-way 75 ft, Egress 20 ft, and cabinet 10 ft. See below Figure 1.



Background

HEFP SEP completed scoping sessions in preparation for the Cerner deployment. HEFP SEP is in the process of finalizing scoping at all sites by November 2021. During these scoping sessions, HEFP SEP validates whether a site has a geographically diverse path to meet the requirement cited above. If the site did not have existing geographically diverse WAN paths, HEFP SEP identified this as a deficiency, included design and construction of the WAN paths in the scope of work and allocated funding in the ROM IGCE the scoping session. The purpose was to ensure the work was included in the scope and provide a magnitude cost estimate to design and construct the secondary conduit run to meet this requirement.

HEFP SEP has discovered that there is currently no national OIT or OEHRM process to coordinate this work with the OIT contracted providers. HEFP SEP is in discussions with OIT and OEHRM to clarify the process of working with OIT contracted providers so that guidance can be provided to the field. Without a clear process to work with OIT contracted providers, it is difficult, and sometimes not possible, to identify the location of the provider's entrance at the property boundary and the resulting pathway to the main service entrances.

Solution

Due to the contractual discussions OIT must have with the WAN providers to coordinate this work, HEFP SEP cannot provide adequate guidance to field engineers to ensure that this requirement is met.

To prevent delay and ambiguity in meeting this requirement, if a VAMC is unable to identify or locate the boundary entrance point, HEFP SEP recommends that the site discontinue this effort in current EHRM NRM projects and de-scope this work from the design and/or construction contracts (whichever applies). At this time, HEFP SEP will not provide updated IGCEs, but VAMCs may adjust cost estimates, as they find appropriate.

If a site has successfully coordinated this work with OIT and included diverse pathways in their designs, please reach out directly to James Cullum and Mike Vulpis so that HEFP SEP can ensure OIT agrees on a case-by-case basis.

When HEFP SEP clarifies the OIT process for implementing this requirement with contracted providers, our office will schedule supplementary scoping sessions with the field to scope this work under a separate EHRM NRM project.

Notes

Please use this link, [VHA HEFP EHRM - 00 VA OEHRM Site Infrastructure Requirements - 2.0 Signed.pdf - All Documents \(sharepoint.com\)](#) to locate the current version of OEHRM Site Infrastructure Requirements – 2.0.

HEFP SEP (SOCAMES 6) Design Alert 7

Problem

There are specific circumstances when construction of a conditioned Telecommunication Room (TR) is infeasible or impractical in non-clinical buildings such as engineering/maintenance buildings, warehouses, temporary modular trailers and quarters that were converted to administrative space. In cases where existing space, structural or other constraints do not practically allow for construction of a TR, the use of a Telecommunication Enclosure (TE) is appropriate.

Background

HEFP SEP has seen a considerable increase in variance requests to deviate from the 80 square foot TR space standard as defined in the *Infrastructure Standards for Telecommunications Spaces* (ISTS). The variance requests specifically address low use, non-clinical TRs where expansion or new construction are not viable solutions due to space and/or environmental constraints.

Solution

The site should make every reasonable effort to comply with the current ISTS standard for all TRs in non-clinical buildings. However, where space and/or environmental constraints cannot support a new TR, both a Standard TE and Non-Standard TE are available for consideration under the conditions described below.

A Variance is NOT REQUIRED for a STANDARD TE (fan-cooled) if all the following criteria are met:

1. **Space Classification:** The space is one of the following:
 - a) Parking structures
 - b) Technical spaces such as warehouses, kitchens, laundries, mechanical/electrical plant buildings, chiller or boiler plants, garages, and paint shops
 - c) Historical quarters converted to administrative use
 - d) Buildings with no VA staffing presence and no requirements for connectivity to the VA LAN or guest Wi-Fi
 - e) Temporary modular trailers. *Temporary* buildings and trailers must be validated as temporary, with a planned removal date and no history of previous deferred or cancelled removal plans.

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2. **Usage:** Usage is limited to a maximum of 96 Work Area Outlets (WAOs). Each data jack in a workspace telecommunications outlet, wired back to the patch panels in the TE, is considered a WAO for these purposes. Where 1-48 WAOs are planned, a 12RU (half-height) standardized TE is used. Where 49-96 WAOs are planned, a 26RU (full height) standardized TE is used.
3. **Environment:** Spaces meet the environmental envelope conditions for a TR, as described in Figure 19 and Table 24 of the Infrastructure Standard for Telecommunications Spaces. Ambient air conditions in the space are between 41°F-95°F dry bulb, 8-80% RH, and dew point less than 82.4°F, measured at the intake point of the TE.

Standard TE's (or equal) are provided below:

12RU - <https://www.chatsworth.com/en-us/products/cabinets-enclosures-containment/wall-mount-cabinets/swing-frame/cube-it-wall-mount-cabinet/11900-e24>

26RU - <https://www.chatsworth.com/en-us/products/cabinets-enclosures-containment/wall-mount-cabinets/swing-frame/cube-it-wall-mount-cabinet/12419-e48>

If all the above criteria are met, except for paragraph 3. *Environment*, a NON-STANDARD (Environmentally Conditioned) TE may be considered, and a Variance is REQUIRED. To request a NON-STANDARD TE Variance:

1. Determine if another space in the building meets the environmental and physical security criteria for the installation of a standardized TE. If so, a standardized TE shall be relocated to the environmentally appropriate space in the building.
2. If no other spaces in the building meet the environmental and physical security criteria, submit a Variance request to DCIE for installation of a non-standard (environmentally conditioned) TE. Select which non-standard TE is proposed for use and submit the installation plan, showing the location on an as-built drawing.
3. A sampling of vendors that may meet the intent of a non-standard (environmentally conditioned) TE solution include:
 - a. Liebert (Vertiv) MCR Mini Computer Room Enclosure
<https://www.vertiv.com/en-us/products-catalog/facilities-enclosures-and-racks/integrated-solutions/mcr---mini-computer-room-enclosure/>
 - b. Hoffman Spectracool – A side-mount A/C unit capable of attaching to many common manufacturer's cabinets (comes in a variety of sizes)
https://hoffman.nvent.com/sites/g/files/hdkjer316/files/acquiadam/2021-05/Spec-00728.pdf?asset_type=Spec

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- c. EIC Solutions Protector Series, Wall-Mount Enclosure S723616 Wall-mount AC Enclosure - 72"H x 36"W x 16"D with EIC Solutions CB Series, Compressor-Based, Side-Mount, Hazardous Location Enclosure CBIQ6000V16XP2 - 6,000 BTU Hazardous Location Air Conditioner <https://www.eicsolutions.com/protector-series-air-conditioned-electronic-enclosures/> and <https://www.eicsolutions.com/product/cbiq6000v16xp2/>

Notes

Variance request information is in the [*Infrastructure Standard for Telecommunication Spaces \(ISTS\)*](#) standard, Section 1.3.1 and Appendix A.

Once approved, the designer of record **MUST** incorporate the approved TE into the design documents.

HEFP SEP (SOCAMES 6) Design Alert 8

Problem

A conflict was noted between the [Infrastructure Standard for Telecommunication Spaces \(ISTS\) v3.1](#) and the [HVAC Design Manual](#) concerning design parameters for temperature and relative humidity ranges in Telecommunication Spaces.

Background

This Design Alert provides temporary guidance on temperature and relative humidity design ranges for Telecommunication Spaces. This Design Alert will remain in effect until the Infrastructure Standard for Telecommunications Spaces and the HVAC Design Manual publish updates that reconcile the temperature and relative humidity parameters.

Solution

The HVAC Design Manual requires a dry bulb temperature of 64° F for heating and 81° F for cooling and a relative humidity of 30% RH to 60% RH for Telecommunication Spaces, (pp. 6-106 for Data Centers and pp. 6-109 for TRs). These temperatures are supply temperatures. The ISTS requires rack inlet temperatures in telecommunications rooms between 41°F and 95°F and between 72°F and 80.6°F in computer rooms with a relative humidity range of 8% to 80%. Please use the [Computational Fluid Dynamics \(CFD\) Modeling Guide](#) for CRAC and rack inlet temperature design in Data Centers while meeting the more stringent of the two standards. To summarize, AC supply temperature set points must be between 64°F and 81°F per the HVAC Design Manual while maintaining rack inlet temperatures between 72°F and 80.6°F in computer rooms. Relative humidity in all telecommunications spaces will be maintained at the more stringent HVAC Design Manual requirements of 30% to 60%.

HEFP SEP (SOCAMES 6) Design Alert 9

Problem

There are specific circumstances when construction of a conditioned Telecommunication Room (TR) is infeasible or impractical in non-clinical buildings such as engineering/maintenance buildings, warehouses, temporary modular trailers, and quarters converted to administrative space. In cases where existing space, structural conditions, or other constraints do not practically allow for construction of a TR, the use of a Telecommunication Enclosure (TE) is appropriate, [See Design Alert 7](#). However, the [Physical Security and Resiliency Design Manual](#) (PSRDM) does not provide clear direction for the physical security of a TE.

Background

When a TE is appropriate, physical security of the enclosure is required. Due to the nature of the enclosure, it is not possible to provide the same physical security required by the PSRDM for Telecommunication Rooms, Section 9.4.3 and [Appendix B \(Security System Matrix\)](#). In this instance, the TE must meet the same standards as the building in which it is placed. For example, if the TE is located within a building classified as Life Safety Protected in accordance with the PSRDM, then the electronic security of the TE is to be protected to that same level (e.g., keyed access, door position switch, video monitoring), to the maximum extent possible (i.e., PIV readers for TEs are not available). Layered security measures must be in place to ensure the TE meets PSRDM criteria to the greatest extent possible.

Solution

The site should make every reasonable effort to comply with the PSRDM, Appendix B for all TEs as if it were a TR. Use the following requirements below as guidance.

Requirements:

- TEs must be in areas not readily accessible to general foot traffic, conditioned, away from high heat sources and can be secured, either by key lock or card reader such as an office, break room, copier room, etc. If in doubt regarding the proposed location, submit a [Station Inquiry](#).
- Comply with the power and cooling requirements for a TE as described in the [Design Alert 7 - Telecommunications Enclosure Design Guide](#).
- Secure the door(s) of the TE with a door position switch, connected to the Physical Access Control System (Appendix B, 10.4 PACS & 10.5 IDS).
- Use a fixed camera with passive infrared (PIR) supplemental lighting (Appendix B, 10.6 SSTV).
 - Note: Due to issues regarding privacy and union contractual requirements, the use of digital masking is required to ensure that only the TE is recorded in the field of view (FOV).
- Incorporate the TE keys into the existing key management program.

HEFP SEP (SOCAMES 6) Design Alert 11

Problem

Some Stations are acquiring incompatible power cordsets for IT devices in Telecommunication Rooms (TRs). Certain networking equipment requires a specific configuration. This Design Alert clarifies selection criteria for power cordsets which connect Rack Power Distribution Units (PDUs) to IT to access-layer network devices resident in TRs.

Additionally, sites shall verify IT equipment Power Supply Unit (PSU) compatibility with 208VAC input voltage before connecting power to prevent possible damage to equipment.

Background

VA OIT power distribution topology leverages vertical Rack PDUs for modularity. The topology also increases the expected circuit voltage delivered to IT devices from 120VAC to 208VAC. The Infrastructure Standard for Telecommunications Spaces (ISTS) v3.1 lists the most common device power cordsets within data centers (Section 4.2.7, Implementation Guidance and Figure 11), although device-specific scenarios are not described. **Cordsets** consist of two couplers and a cable manufactured as single unit. A specific cordset configuration is necessary to support access-layer networking equipment in TRs.

Enterprise-grade IT devices use paired couplers for power delivery, allowing the use of detachable country-specific power cordsets. These coupler variations are defined by the International Electrotechnical Commission (IEC) in their IEC 60320 standard. Typically, **even-numbered couplers** hold exposed blades within a recess, and the blades point towards the power source; and **odd-numbered couplers** fit in the recess and receive the blades. According to IEC 60320, **plugs** on cordsets are paired with and provide power downstream to devices through recessed blades in **Inlet** housings; interconnection **Outlets** provide power from one device to another device through cordsets with bladed **Plug Connectors**. This Design Alert will use those terms; however, vendor descriptions of the necessary couplers may not adhere to exact terminology used in IEC 60320. Ambiguous descriptions should be visually confirmed through pictures or drawings. Stations may submit an HEFP-SEP EHRM Station Inquiry for assistance with situations not addressed in this Design Alert.

Rack PDUs are considered intermediate devices in IEC 60320 coupler architecture, because they both receive and provide power. Thus, Rack PDU outlet banks are populated with IEC 60320 C13 Outlets. Devices supported from C13 Outlets require cordsets with matching C14 Plug Connectors (Figure 1).

Access-layer networking equipment PSUs use C16 Inlet couplers rated for elevated temperature environments up to 120°C (248°F). Similarly rated C15 Plugs are required at the device end of the cordset (Figure 2). By design, C16 Inlets are specifically keyed to prevent insertion of lower temperature rated C13 Plugs, although C14 Inlets will accept both C13 and C15 Plugs (Figure 3).

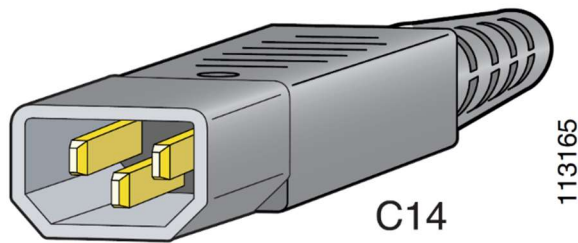


Figure 1: IEC 60320 C14 Plug Connector.
Source: Cisco Systems Inc. Illustration #113165

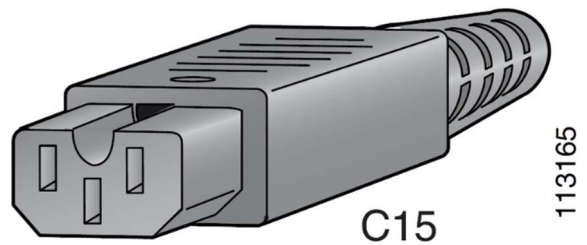


Figure 2: IEC 60320 C15 Plug.
Source: Cisco Systems Inc. Illustration #113165

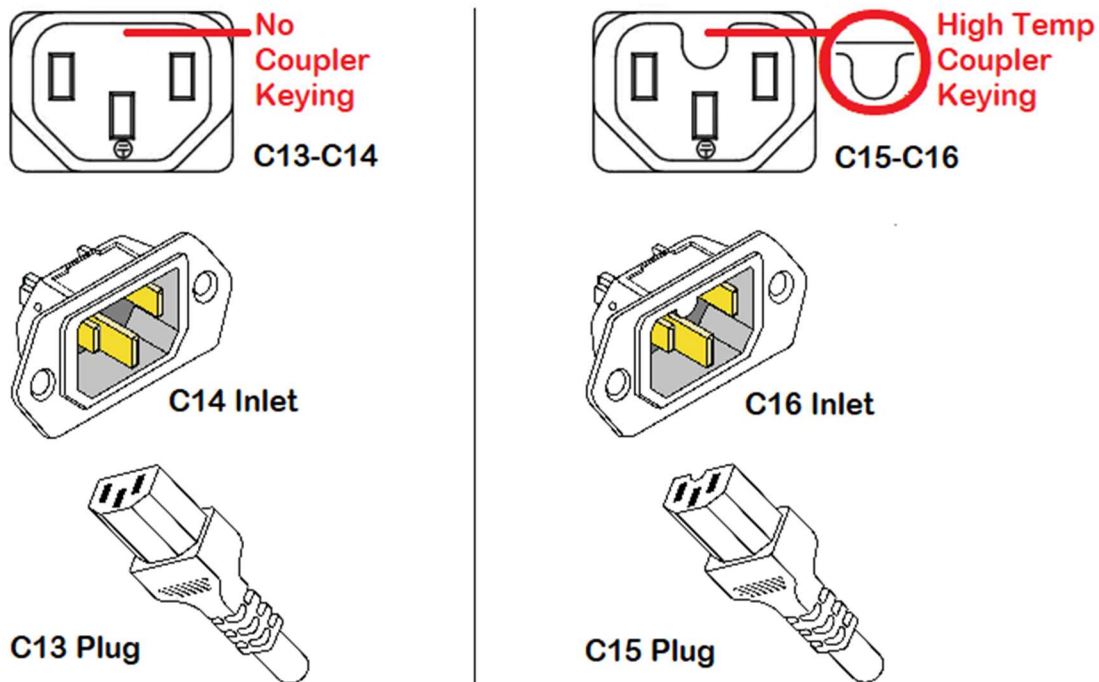


Figure 3: Keying difference between IEC 60320 C13-C14 Couplers and IEC 60320 C15-C16 Couplers.

Additional care must be taken when converting from 120VAC power distribution to 208VAC power distribution within a space. **Installers must confirm that every cordset and every device is compatible with the source power voltage before connecting them, otherwise equipment damage may result.**

Although IEC 60320 requires cable voltage ratings up to 250VAC, localized cordsets which may only be rated for 125VAC have been manufactured to support consumer devices. Voltage ratings for cordsets should be imprinted along the cable jacket or on the coupler ends. Even if a cordset's couplers are compatible with the Rack PDU Outlets and the device Inlet, cordsets rated for only 120VAC should not be used with 208VAC distribution.

Installers must confirm that every cordset, and device is compatible with the source power voltage before connecting them, otherwise equipment damage may result.

Most enterprise-grade IT devices are manufactured with auto-ranging PSUs, which can detect and adjust to local power circuit characteristics. PSU Inlets should be clearly marked with the ranges they can accept. Usual power input ranges cover combinations of 100-125VAC and 200-250VAC as well as 50Hz and 60Hz frequencies. However, other devices with consumer-grade PSUs which accepts only 120VAC and cannot accept 208VAC power may be encountered in Telecommunications Rooms. These device PSUs will generally have C14 Inlets, so there may be no physical barrier to prevent inadvertent connection of 208VAC cordset couplers to 120VAC rated Inlets.

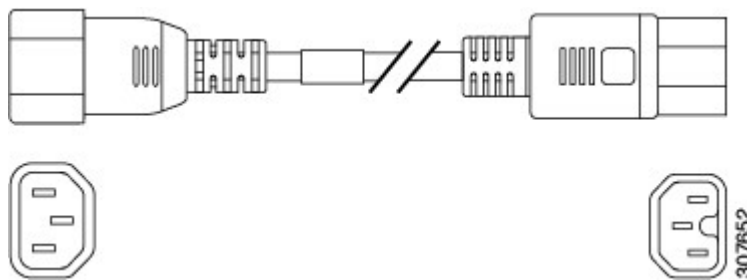
Installers must confirm that every cordset, and every device is compatible with the source power voltage before connecting them, otherwise equipment damage may result.

Solution

In TRs where power distribution has been updated to use 208VAC, specify and use IEC 60320 C14-C15 device power cordsets to connect access-layer networking equipment in TR racks with 208V Rack PDU Outlets.

IEC 60320 C14-C15 device power cordsets should be functionally equivalent to Cisco Systems Cable Part **CAB-C15-CBN** for commonality and compatibility (Figure 4).

- Rated for at least 250 Volts AC, 13 Amps, 60 Hz, 120°C (248°F) at coupler pins
- Underwriters Laboratories (UL) listed
- 4 ft (1.22m) maximum overall length; shorter lengths may be used to reduce excessive slack
- Coupler terminations on cordset:
 - IEC 60320 C14 Plug Connector (Rack PDU End)
 - IEC 60320 C15 Plug (Device End)
- Selection of contrasting cordset colors indicating diverse (A/B) power distribution sources is recommended.



*Figure 4: Cisco IEC C14-C15 Cabinet Jumper Power Cord, Power Cable Part Number **CAB-C15-CBN**.
Source: Cisco Systems Inc. Illustration #307652*

- Verify cordset compatibility with 208VAC source power before connecting.
- Verify device compatibility with 208VAC input power before connecting.

HEFP SEP (SOCAMES 6) Design Alert 12 Design**Problem**

PG 18-15 Volume C, AE Submission Instructions for Minor and NRM Construction Program [A/E Submission Instructions - Minor and NRM \(va.gov\)](#) does not provide detailed requirements for telecommunication system submissions. As a result, HEFP SEP receives incomplete design packages. - Incomplete design packages are often rejected which result in significant schedule delays.

Background

PG-18-15, Volume C was updated in November 2008 when telecommunications systems were not as complex and robust; therefore, it does not include detailed requirements for telecommunications system submissions. Due to the unique elements that must be addressed in the EHRM designs, HEFP SEP developed detailed telecommunication system submission requirements.

Solution

Include the below document in new AE Scopes of Work and/or distribute to AE firms that are currently in design. This document is limited to HEFP-SEP EHRM NRM projects. This document does not supersede PG-18-15 Volume C or other associated attachments to the A/E's awarded Scope of Work.

Telecommunications:	10%	SD (35%)	DD (65%)	CD (95%)
Narrative Elements:				
XMATRIX Summary Table (From HEFP-SEP Site Self-Assessment)	X	X	X	X
Existing Conditions	X	X	X	X
Assumptions	X	X	X	X
Media Types (copper cabling, fiber (inside plant and outside plant if applicable))	X	X	X	X
Design Description		X	X	X
At-Commissioning Load Levels			X	X
Ultimate Design Load Levels			X	X
Computational Fluid Dynamics (CFD) analysis			X	X
Drawings showing:				
Diamond analysis for horizontal distribution	X	X	X	X
Campus telecommunications pathways		X	X	X
Telecommunications space locations		X	X	X
Templated Layouts for 1, 2, 3, 4 rack TRs (aligned to ISTS)		X	X	X
Internal telecommunications pathways between Entrance Rooms, MCR, and TRs			X	X
Telecommunications space layouts & outfitting (equipment & electrical/HVAC locations, clearances, etc.)			X	X
Telecommunications space elevation drawings (equipment, deconfliction of services, etc.)			X	X
Space & enclosure bonding details			X	X
Cable media schedules (type, quantity, length, termination points, termination type, etc.)			X	X
Structured cabling maps			X	X
Riser diagrams			X	X
Telecommunications equipment details			X	X
Enclosure elevations				X
Cable labeling matrices for cable plant and patch cord media				X
Equipment submittals:				
IT equipment enclosures				X
Fiber & UTP patch panels				X
Cable media				X
Cable media conveyances				X
Power distribution equipment supporting telecommunications spaces (Rack UPS systems, Data Center UPS, busways, Zone PDUs, Rack PDUs, etc.)				X
Heat rejection equipment supporting telecommunications spaces (CRACs, mini-split air conditioners, etc.)				X
Accessories for the above equipment necessary to complete the design				X

HEFP SEP Design Alert 13

Problem

The HEFP SEP scope of work and x-matrix required Telecommunication Room (TR) environmental monitors on each rack, whereas the Site Infrastructure and End User Device Requirements (SIEUDR) only require environmental monitoring within 6 feet of rack.

Background

During site scoping sessions, HEFP SEP provided documents that required environmental monitors on each rack. Further review of the SIEUDR and Infrastructure Standard for Telecommunication Spaces ([ISTS](#)), version 3.1, section 4.3.2 *Environmental Operating Envelope Conditions*, indicate that environmental monitors are only required within 6 feet of the racks in TRs.

Solution

Per [ISTS](#), version 4.0, section 4.3.7 *Monitoring > Implementation Guidance*, “IT equipment rack temperature/humidity sensors are required for Campus Data Centers (CDC), Campus Support Centers (CSC), Medical Support Centers (MSC), where there is separation between cold and hot air streams. Rack-level sensors are not required in TRs or entrance rooms where heat rejection is accomplished using the entire room as a heat sink.” For TRs, a temperature and humidity, wall mounted sensor is required within six (6) feet of the rack ([SIEUDR](#), Section 2.5-A-2). Therefore, rack-level sensors, in addition to the wall mounted sensor, are only required in the data center.

Remove all references to rack-level sensors in TRs for EHRM designs at the 95% design stage or earlier. For completed 95% designs or beyond, including those already submitted to the contracting office for bid, consult with your HEFP SEP Program Engineer to determine impacts as a potential cost reduction or value engineering activity. For facilities currently in construction, work with your HEFP SEP Program Engineer to pursue contract modifications, as needed.

Signature Authority

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Distribution

- | | |
|-----------------------------|--------------------------------|
| HEFP SEP Director | HEFP Program Analyst |
| HEFP SEP Capital Engineers | HEFP SEP Contract Support Team |
| VISN Capital Asset Managers | Station Engineering Staff |

HEFP SEP (SOCAMES 6) Design Alert 14

Problem

A conflict was identified between the [Infrastructure Standard for Telecommunication Spaces \(ISTS\)](#) and the [HVAC Design Manual](#) concerning Heating, Ventilation and Air Conditioning (HVAC) redundancy requirements in Telecommunication Rooms (TRs). The ISTS requires (N) HVAC redundancy for TRs, however the HVAC Design Manual requires (N+1) redundancy for TRs.

Background

This Design Alert provides temporary guidance on HVAC redundancy requirements for TRs in conjunction with [Design Alert 8 – Temp and Humidity Conflict Resolution](#). This Design Alert will remain in effect for HEFP SEP funded EHRM projects until the ISTS and the HVAC Design Manual publish updates that reconcile the HVAC redundancy design parameters.

Solution

EHRM projects shall meet a cooling redundancy requirement of (N) for all Telecommunications Rooms.

HEFP SEP Design Alert 15

Problem

Electronic Health Record Modernization Non-Recurring Maintenance (EHRM NRM) projects include the update of Telecommunication Rooms (TRs) throughout VHA as part of the overall infrastructure upgrades required for EHRM implementation. In general, the EHRM project scope is limited to the boundaries of existing TRs. In many cases, the use of existing space creates challenges in meeting humidity, air change, filtration, and pressure requirements as defined in the VHA HVAC Design Manual. This deficiency is due to limited space and the lack of or limits to existing central air handlers and ductwork.

Background

The VHA HVAC Design Manual requires that facilities comply with all HVAC parameters through connection to central systems or installation of appropriate Computer Room Air conditioner (CRAC) units (with filtration, temperature, and humidity control). However, most existing facility infrastructures was not originally constructed to support today's standards for humidity, filtration, pressurization, and outdoor air in each TR. Due to limited EHRM NRM project budgets and existing space constraints, it is understood that in some situations it is impractical or impossible for facilities to comply.

Adjacent spaces must be evaluated for the possibility of accommodating the appropriate HVAC equipment to serve the TR room prior to considering alternatives to or deviations from the HVAC Design Manual requirements. In some cases, existing system and/or space modifications necessary to comply with the VHA HVAC Design Manual requirements for humidity control ranges, filtration, pressurization, and outside air supply are so significant that they are outside the approved scope of work and budget for the EHRM NRM project. In these cases, alternate HVAC equipment such as mini-split systems may be approved as an interim solution to ensure minimum requirements for the Electronic Health Record deployment are met.

Solution

Since VHA HEFP SEP currently manages the review and implementation of EHRM NRM projects, VHA HEFP OHE delegated authority to the VHA HEFP SEP, to assess and adjudicate deviations to the HVAC Design Manual standard for TRs impacted by the deployment of the EHRM initiative, subject to the below constraints:

1. Approved deviations are abeyances only and must include a detailed and time-defined plan to bring the space(s) into compliance with HVAC Design Manual within 8 years.
2. Facilities must submit deviations to SEP for review through the Station Inquiry (SI) Tool. TRs must be evaluated individually to assess space constraints and infrastructure impacts. Unique circumstances (e.g., adjacent spaces

suitable for HVAC infrastructure support, HVAC availability and capacity, ceiling heights, locations on exterior walls) will differ for each TR, requiring individual evaluation and determinations. Facilities shall ensure every effort to identify a solution which maintains compliance with the HVAC Design Manual prior to requesting an abeyance.

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Distribution

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HEFP SEP (SOCAMES 6) Design Alert 16

Problem

Healthcare Environment and Facilities Programs (HEFP) Special Engineering Projects (SEP) has identified that VA Medical Centers (VAMC) are over provisioned with the number of data drops and cable pulls that currently exist. Therefore, the Work Area Outlet (WAO) count should be based on the Infrastructure Standard Telecommunications Spaces (ISTS) standard of active port count with appropriate provision for additional capacity.

Over-provisioning results in significant unnecessary costs for additional Telecommunication Room (TR) racks and cooling to support both those passive and active ports.

Background

Many Architectural-Engineering (A/E) firms have been designing using a 1:1 replacement scenario of the WAO in rooms; however, ISTS states in Section 4.4.9, that the typical standard density is a 2-port WAO. SEP identified that the overprovisioned planning and design results in a significant percentage of passive and unnecessary drops. The 1:1 replacement scenario resulted in preventable changes; the extraneous number of ports leads to more cable, conduit, racks, patch panels, HVAC, power and most importantly, impacts to Veteran care during construction.

To correct the number of port counts and reduce cost and construction impact, SEP is implementing the below solution.

Note: ISTS requires two+ racks per TR supporting clinical areas regardless of switch counts. However, this design alert provides an alternative design solution.

Solution

1. If at 35% design or earlier:
 - a. A/E to assess all spaces for WAO current state.
 - b. Assess number of cables in each conduit to each WAO.
 - c. Site to validate WAO current state and determine if passive ports are required.
 - i. If there are more than two drops per WAO, reduce to two drops unless High Density is justified.
 - ii. Two Category 6A cables fit in $\frac{3}{4}$ " conduit with 40% fill capacity. This allows reuse of existing conduit and WAOs with minimal construction and time disruption.
 - d. A/E to revise drawings to reflect site determined reductions.
 - e. Site to obtain active "Port Count Report" from SEP.

- i. The number of active ports determines number of active switches required. The active switch count is seen on the Port Count Report in the column titled “new switch count”.
 - ii. As national Office of Information & Technology (OI&T) refreshes the switches, OI&T will limit the number of switches provided based on number of active port counts.
 - iii. In some cases, the switch refresh may have already occurred at site.
- f. The number of switches will determine the number of racks required in a TR.
 - i. 1-3 switches will require 1 rack.
 - ii. 4-8 switches will require two racks.
 - iii. 9+ switches will require three+ racks.
- g. A/E should follow Design Alert 4 ([Design Alerts](#)).
 - i. The power, UPS and cooling should be sized based on rack count.
- h. If only one rack is required by steps above, then:
 - i. Allocate space for 2nd rack and show it as future purchase on design drawings: Not in Contract (N.I.C).
 - ii. Include appropriate power for **two racks** (in case future expansion is needed).
 - iii. Include HVAC for **one rack** with space identified for additional cooling in future.

2. If beyond 35% design -

- a. A/E to assess all spaces for WAO current state.
 - i. Assess number of cables in each conduit to each WAO.
- b. Site to validate WAO current state and determine if passive ports are required.
 - i. If there are more than two drops per WAO, reduce to two drops unless High Density is justified.
 - ii. Two Category 6A cables fit in ¾” conduit with 40% fill capacity. This allows reuse of existing conduit and WAOs with minimal construction and time disruption.
- c. A/E to revise drawings to reflect site determined reductions.
- d. Site to obtain active “Port Count Report” from SEP.
 - i. The number of active ports determines number of active switches required. The Active switch count is seen on the Port Count Report in the column titled “new switch count”.
 - ii. As national OI&T refreshes the switches, OI&T will limit the number of switches provided based on number of active port counts.
 - iii. In some cases, the switch refresh may have already occurred at site.
- e. The number of switches will determine the number of racks required in a TR.
 - i. 1-3 switches will require 1 rack.
 - ii. 4-8 switches will require two racks.
 - iii. 9+ switches will require three+ racks.
- f. If A/E shows larger rooms and they are approved, keep them as sized, but:
 - i. Allocate space for 2nd rack and show it as future purchase on design drawings: Not In Contract (N.I.C).
 - ii. Include appropriate power for **two racks** (in case future expansion is needed).
 - iii. Include HVAC for **one rack** with space identified for additional cooling in future.

3. If in construction –

- a. Site to validate WAO current state against drawings and determine if passive ports are required.

- i. Consider number of cables in each conduit to each WAO.
 - ii. If there are more than two drops per WAO, consider reducing to two.
 - iii. Two Category 6A cables fit in $\frac{3}{4}$ " conduit with 40% fill capacity. This allows reuse of existing conduit and WAOs with minimal construction and time disruption.
 - iv. **Assess number of cables General Contractor (GC) is pulling in each conduit to each WAO verses existing conditions.**
 - 1. **If pulling more than two Category 6A cables to every WAO, consult with SEP.**
 - 2. **If site is replacing all existing conduit in the walls, consult with SEP.**
 - b. Site to obtain active "Port Count Report" from SEP.
 - i. The number of active ports determines number of active switches required. The Active switch count is seen on the Port Count Report in the column titled "new switch count".
 - ii. As national OI&T refreshes the switches, OI&T will limit the number of switches provided based on number of active port counts.
 - iii. In some cases, the switch refresh may have already occurred at site.
 - c. The number of switches will determine the number of racks required in a TR.
 - i. 1-3 switches will require 1 rack.
 - ii. 4-8 switches will require two racks.
 - iii. 9+ switches will require three+ racks.
 - iv. If GC has not purchased racks, issue a modification to reduce rack count based on above.
 - v. If GC has not purchased supporting HVAC, issue a modification for reduction of cooling based on rack count above.
4. It is important for the A/E and site, while reviewing the Port Count Report, to be cognizant of the following:
- a. Room numbers on the report may not match what is space approved.
 - i. Switch stack naming conventions include Building and Room numbers within the names.
 - ii. Site room numbering may not match exactly. We have seen differences in the existing room labels, space files and the Port Count Report.
 - iii. Switches may need to be split into multiple TRs or consolidated into TRs. The A/E should plan accordingly.
 - b. The Port Count Report only accounts for OI&T switches on the network.
 - c. If there is a SWO1 and SWO2 with same building and room nomenclature, they are two racks that are probably in the same TR. The new TR will need to count both switch counts.


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